CO_2 Emission Prediction Case Study

Business Objective

The Global Automotive Council aims to understand the key factors influencing vehicle CO_2 emissions and identify opportunities for emission reduction.

This analysis explores vehicle specifications, engine characteristics, transmission systems, fuel types, and fuel consumption patterns to uncover factors that drive emissions.

The study includes:

- Data quality assessment and preprocessing
- Exploratory Data Analysis (EDA)
- Feature relationship analysis
- Vehicle and fuel category comparisons
- Outlier investigation
- Predictive modeling using regression
- Model evaluation and interpretation
- Business recommendations for emission reduction

Target Variable: **CO_2 Emissions (g/km)**

In [1]:
```
import pandas as pd
import numpy as np
```

In [2]:
```
url = "https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/000/158/208/original/CO2_Emissions.csv?1760012266"

df = pd.read_csv(url)
df.to_csv("carbon_emission_dataset.csv", index=False)
```

1. Dataset Overview

The dataset contains information about vehicle specifications, fuel consumption characteristics, and corresponding CO_2 emissions.

The objective is to understand how vehicle design and operational characteristics influence carbon emissions and to build a model capable of estimating CO_2 emissions based on these features.

In [4]:
```
print("Dataset Shape :", df.shape)

df.info()
```

```
Dataset Shape : (7385, 12)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7385 entries, 0 to 7384
Data columns (total 12 columns):
 #   Column                            Non-Null Count  Dtype
---  ------                            --------------  -----
 0   Make                              7385 non-null   object
 1   Model                             7385 non-null   object
 2   Vehicle Class                     7385 non-null   object
 3   Engine Size(L)                    7385 non-null   float64
 4   Cylinders                         7385 non-null   int64
 5   Transmission                      7385 non-null   object
 6   Fuel Type                         7385 non-null   object
 7   Fuel Consumption City (L/100 km)  7385 non-null   float64
 8   Fuel Consumption Hwy (L/100 km)   7385 non-null   float64
 9   Fuel Consumption Comb (L/100 km)  7385 non-null   float64
 10  Fuel Consumption Comb (mpg)       7385 non-null   int64
 11  CO2 Emissions(g/km)               7385 non-null   int64
dtypes: float64(4), int64(3), object(5)
memory usage: 692.5+ KB
```

In [5]:
```
df.head()
```

Out[5]:

	Make	Model	Vehicle Class	Engine Size(L)	Cylinders	Transmission	Fuel Type	Fuel Consumption City (L/100 km)	Fuel Consumption Hwy (L/100 km)	Fuel Consumption Comb (L/100 km)	Fuel Consumption Comb (mpg)	C Emissions(g/k
0	ACURA	ILX	COMPACT	2.0	4	AS5	Z	9.9	6.7	8.5	33	1
1	ACURA	ILX	COMPACT	2.4	4	M6	Z	11.2	7.7	9.6	29	2
2	ACURA	ILX HYBRID	COMPACT	1.5	4	AV7	Z	6.0	5.8	5.9	48	1
3	ACURA	MDX 4WD	SUV - SMALL	3.5	6	AS6	Z	12.7	9.1	11.1	25	2
4	ACURA	RDX AWD	SUV - SMALL	3.5	6	AS6	Z	12.1	8.7	10.6	27	2

In [6]:
```
df.describe().T
```

Out[6]:

	count	mean	std	min	25%	50%	75%	max
Engine Size(L)	7385.0	3.160068	1.354170	0.9	2.0	3.0	3.7	8.4
Cylinders	7385.0	5.615030	1.828307	3.0	4.0	6.0	6.0	16.0
Fuel Consumption City (L/100 km)	7385.0	12.556534	3.500274	4.2	10.1	12.1	14.6	30.6
Fuel Consumption Hwy (L/100 km)	7385.0	9.041706	2.224456	4.0	7.5	8.7	10.2	20.6
Fuel Consumption Comb (L/100 km)	7385.0	10.975071	2.892506	4.1	8.9	10.6	12.6	26.1
Fuel Consumption Comb (mpg)	7385.0	27.481652	7.231879	11.0	22.0	27.0	32.0	69.0
CO2 Emissions(g/km)	7385.0	250.584699	58.512679	96.0	208.0	246.0	288.0	522.0

2. Data Quality Assessment

Before analysis, the dataset is examined for:

- Missing values
- Duplicate records
- Incorrect data types
- Potential inconsistencies

This ensures that subsequent analysis is reliable and meaningful.

In [7]:
```
df.isnull().sum()
```

Out[7]:
```
Make                              0
Model                             0
Vehicle Class                     0
Engine Size(L)                    0
Cylinders                         0
Transmission                      0
Fuel Type                         0
Fuel Consumption City (L/100 km)  0
Fuel Consumption Hwy (L/100 km)   0
Fuel Consumption Comb (L/100 km)  0
Fuel Consumption Comb (mpg)       0
CO2 Emissions(g/km)               0
dtype: int64
```

In [8]:
```
print("Duplicate Rows :", df.duplicated().sum())
```

```
Duplicate Rows : 1103
```

In [9]:
```
df.dtypes
```

Out[9]:
```
Make                              object
Model                             object
Vehicle Class                     object
Engine Size(L)                   float64
Cylinders                          int64
Transmission                      object
Fuel Type                         object
Fuel Consumption City (L/100 km)  float64
Fuel Consumption Hwy (L/100 km)   float64
Fuel Consumption Comb (L/100 km)  float64
Fuel Consumption Comb (mpg)        int64
CO2 Emissions(g/km)                int64
dtype: object
```

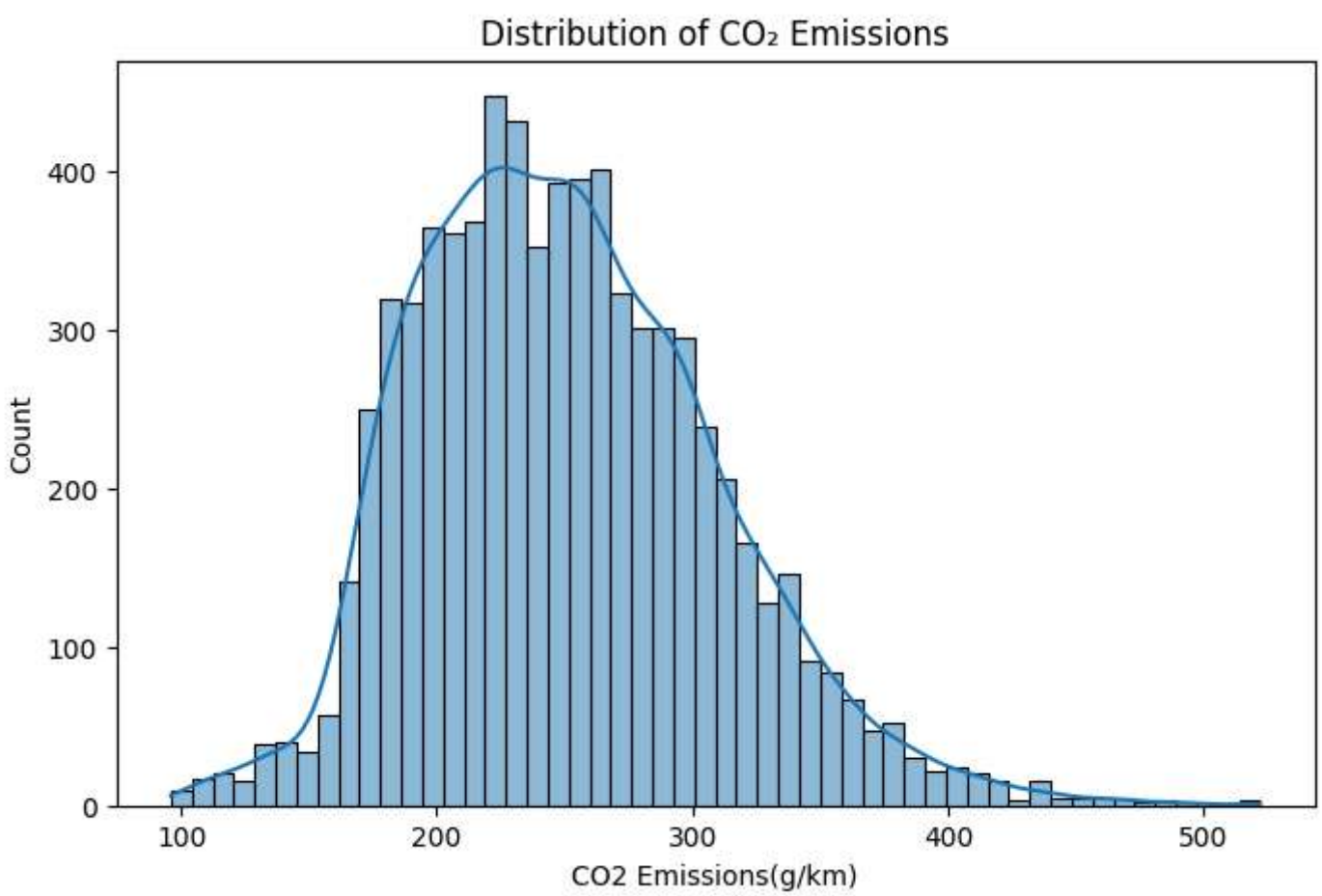
3. CO_2 Emission Distribution Analysis

Understanding the distribution of CO_2 emissions helps identify skewness, variability, and potential outliers.

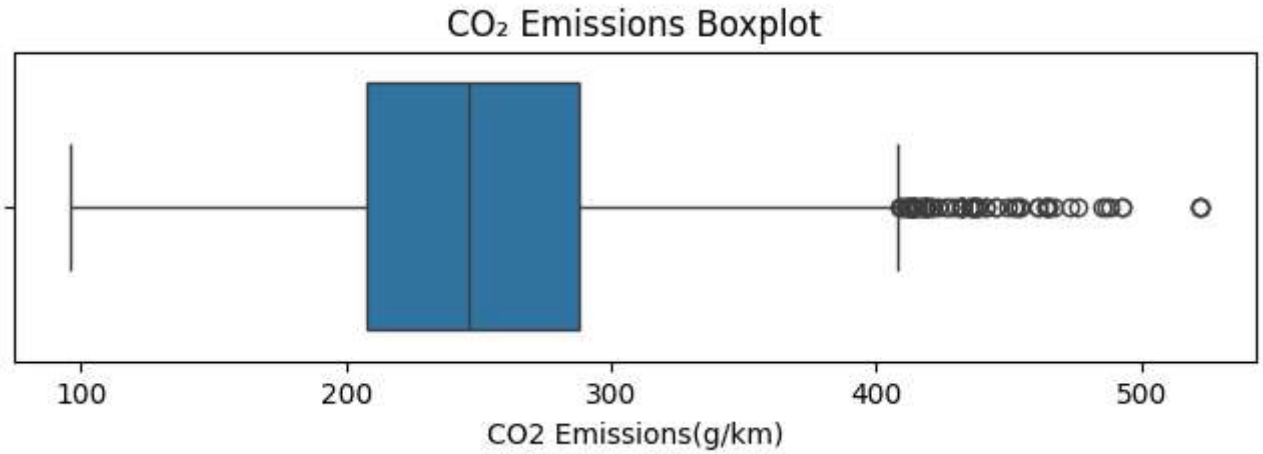
Histograms and boxplots are used to evaluate the overall emission pattern.

In [10]:
```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8,5))
sns.histplot(df["CO2 Emissions(g/km)"], kde=True)
plt.title("Distribution of CO₂ Emissions")
plt.show()
```



In [12]:
```
plt.figure(figsize=(8,2))
sns.boxplot(x=df["CO2 Emissions(g/km)"])
plt.title("CO₂ Emissions Boxplot")
plt.show()
```



4. Numerical Feature Analysis

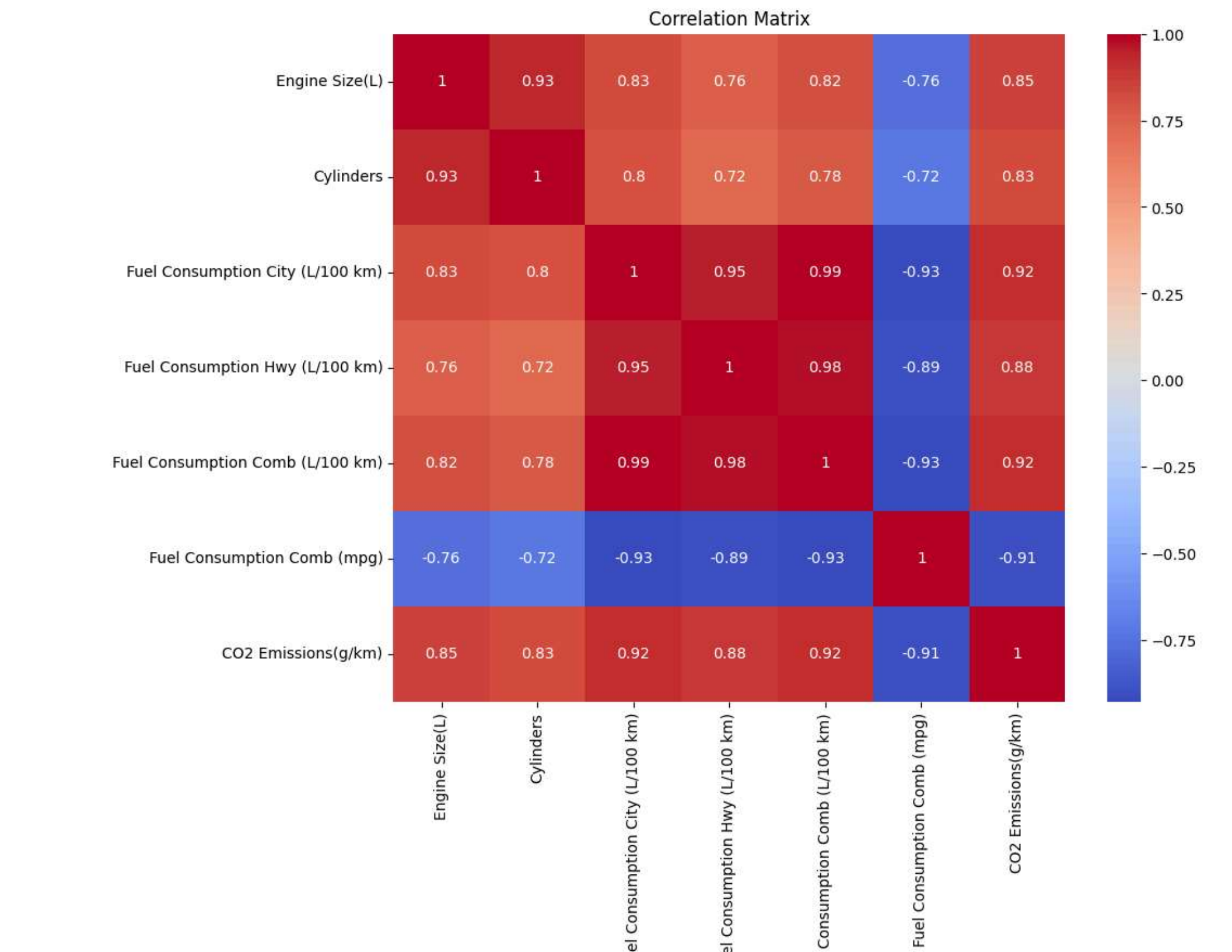
The relationship between numerical vehicle characteristics and CO_2 emissions is examined using correlation analysis.

Strong correlations indicate variables that may significantly influence emissions.

In [13]:
```
num_cols = df.select_dtypes(include=np.number).columns

corr = df[num_cols].corr()

plt.figure(figsize=(10,8))
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Matrix")
plt.show()
```


Correlation Matrix
Engine Size(L)
Cylinders
Fuel Consumption City (L/100 km)
Fuel Consumption Hwy (L/100 km)
Fuel Consumption Comb (L/100 km)
Fuel Consumption Comb (mpg)
CO2 Emissions(g/km)
1 0.93 0.83 0.76 0.82 -0.76 0.85
0.93 1 0.8 0.72 0.78 -0.72 0.83
0.83 0.8 1 0.95 0.99 -0.93 0.92
0.76 0.72 0.95 1 0.98 -0.89 0.88
0.82 0.78 0.99 0.98 1 -0.93 0.92
-0.76 -0.72 -0.93 -0.89 -0.93 1 -0.91
0.85 0.83 0.92 0.88 0.92 -0.91 1
Engine Size(L)
Cylinders
el Consumption City (L/100 km)
el Consumption Hwy (L/100 km)
Consumption Comb (L/100 km)
Fuel Consumption Comb (mpg)
CO2 Emissions(g/km)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75

Fu Fu Fuel

In [14]:
```
corr["CO2 Emissions(g/km)"].sort_values(ascending=False)
```

Out[14]:
```
CO2 Emissions(g/km)                1.000000
Fuel Consumption City (L/100 km)   0.919592
Fuel Consumption Comb (L/100 km)   0.918052
Fuel Consumption Hwy (L/100 km)    0.883536
Engine Size(L)                     0.851145
Cylinders                          0.832644
Fuel Consumption Comb (mpg)       -0.907426
Name: CO2 Emissions(g/km), dtype: float64
```

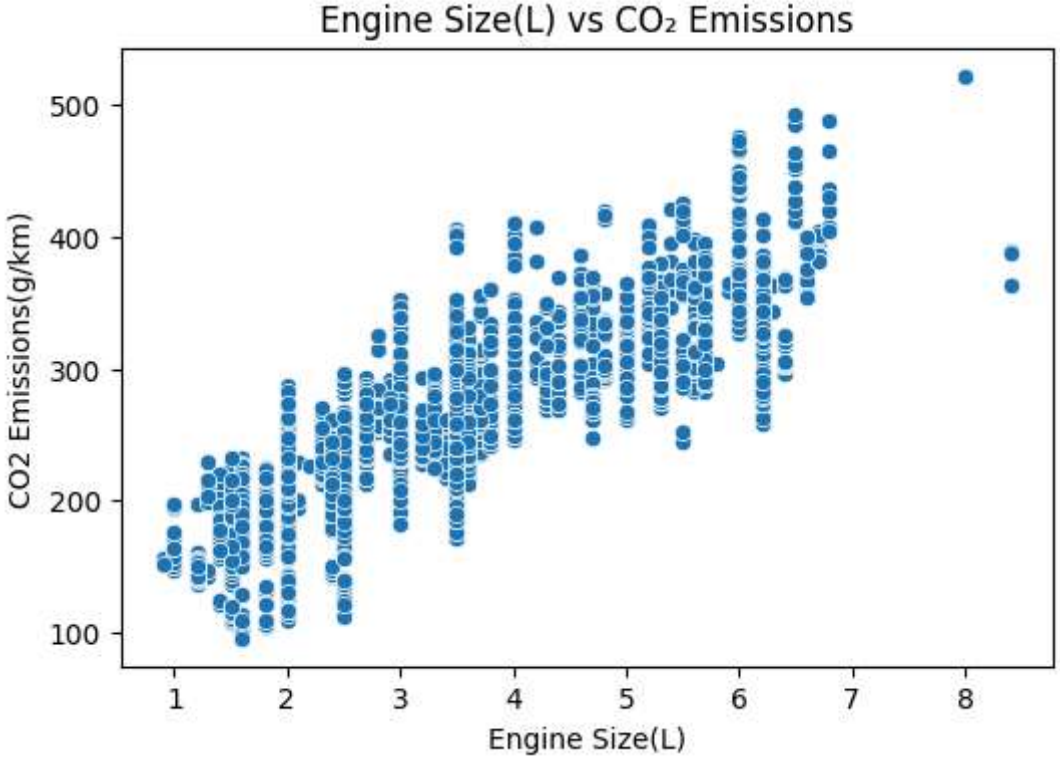
5. Emission Trends Across Numerical Variables

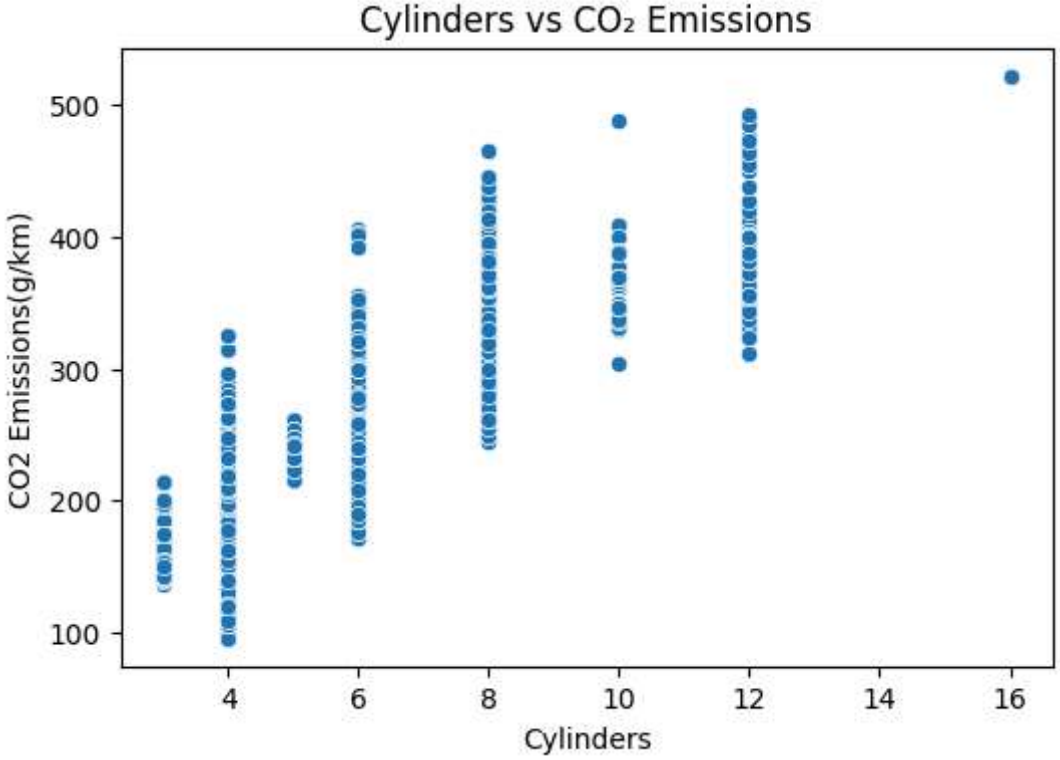
Scatter plots help visualize how CO_2 emissions change with key numerical vehicle characteristics.

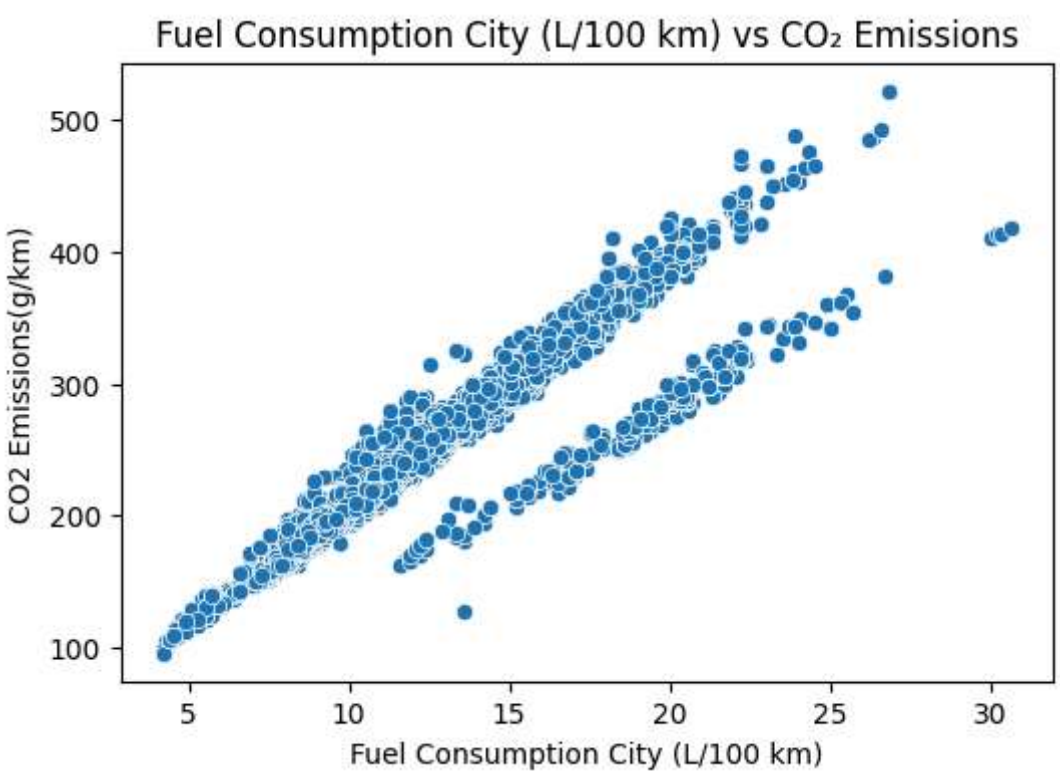
These plots reveal whether relationships appear linear, non-linear, or influenced by outliers.

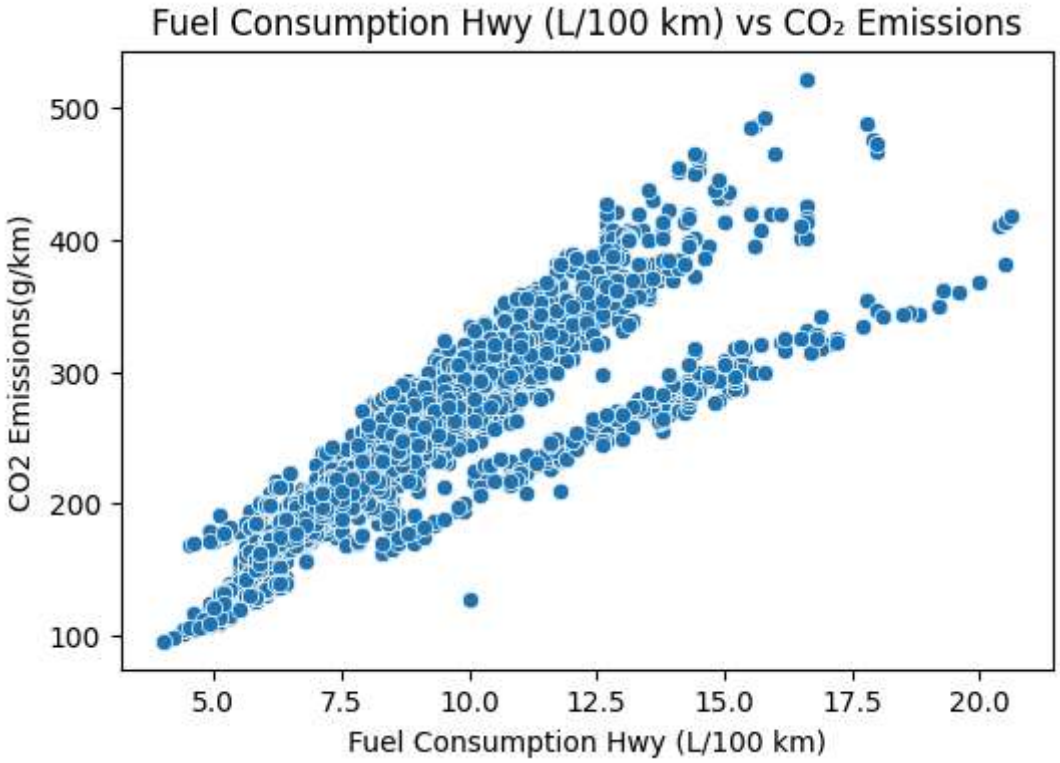
In [15]:
```
num_features = [
    "Engine Size(L)",
    "Cylinders",
    "Fuel Consumption City (L/100 km)",
    "Fuel Consumption Hwy (L/100 km)",
    "Fuel Consumption Comb (L/100 km)"
]

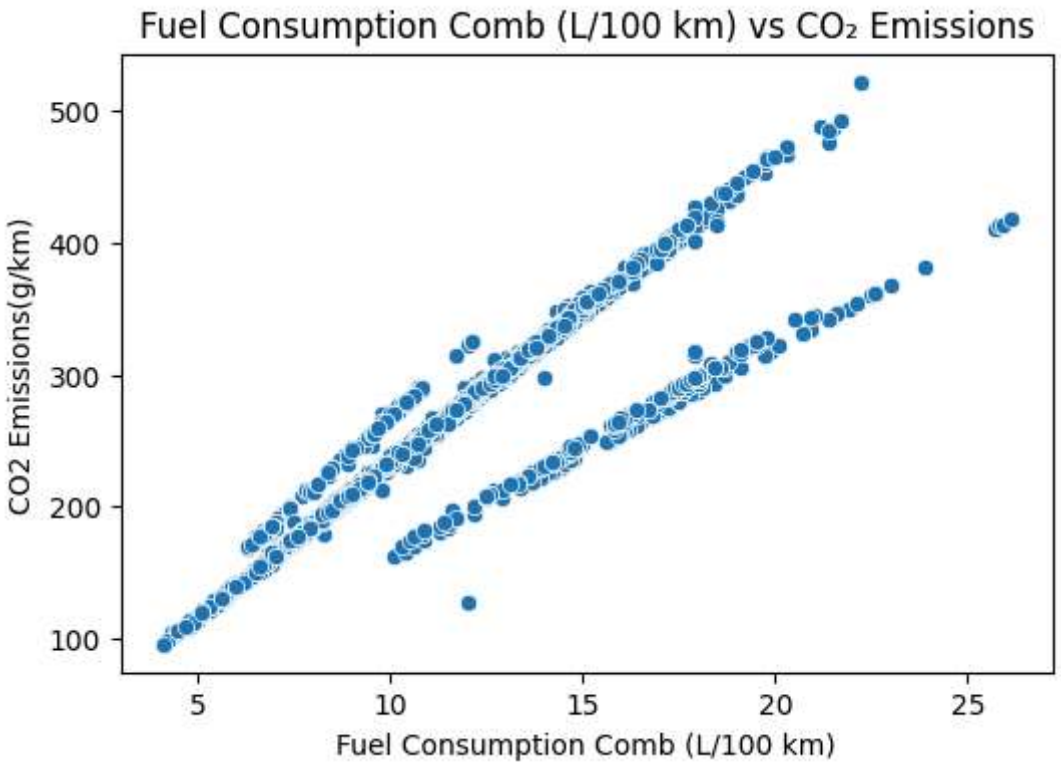
for col in num_features:
    plt.figure(figsize=(6,4))
    sns.scatterplot(data=df,
                    x=col,
                    y="CO2 Emissions(g/km)")
    plt.title(f"{col} vs CO₂ Emissions")
    plt.show()
```


Engine Size(L) vs CO₂ Emissions
CO2 Emissions(g/km)
Engine Size(L)
100
200
300
400
500
1
2
3
4
5
6
7
8


Cylinders vs CO₂ Emissions
CO2 Emissions(g/km)
Cylinders
100
200
300
400
500
4
6
8
10
12
14
16


Fuel Consumption City (L/100 km) vs CO₂ Emissions
CO2 Emissions(g/km)
100
200
300
400
500
5
10
15
20
25
30
Fuel Consumption City (L/100 km)


Fuel Consumption Hwy (L/100 km) vs CO₂ Emissions
CO2 Emissions(g/km)
100
200
300
400
500
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Fuel Consumption Hwy (L/100 km)



Observation

A strong positive relationship is observed between fuel consumption measures and CO_2 emissions.

Vehicles with larger engines and more cylinders generally exhibit higher fuel consumption and consequently higher emissions.

6. Vehicle Class Comparison

Vehicle classes differ in weight, engine capacity, and intended usage.

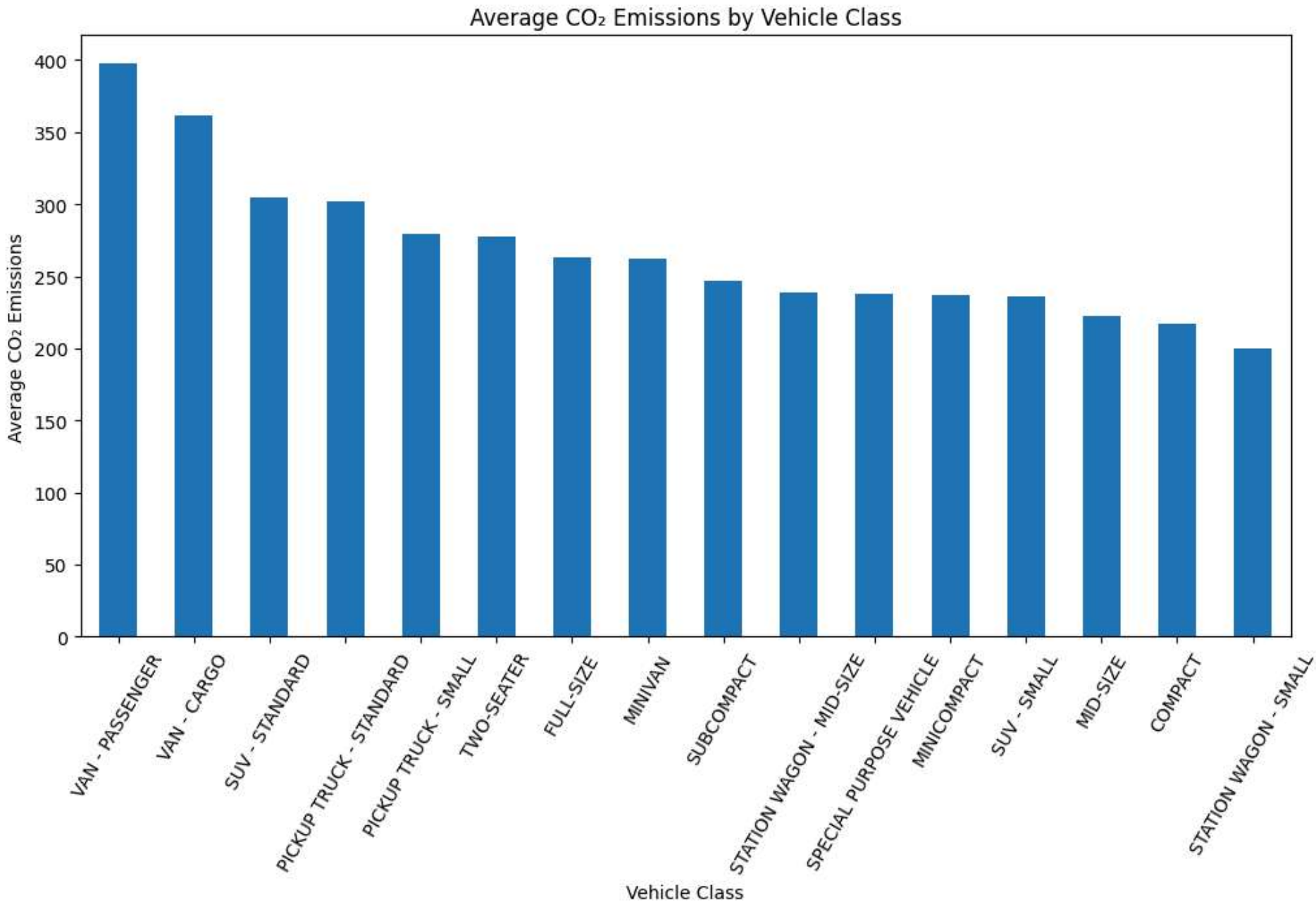
This analysis compares average CO_2 emissions across vehicle categories.

In [16]:
```
vehicle_emissions = (
    df.groupby("Vehicle Class")["CO2 Emissions(g/km)"]
    .mean()
    .sort_values(ascending=False)
)

plt.figure(figsize=(12,6))
vehicle_emissions.plot(kind="bar")
plt.title("Average CO₂ Emissions by Vehicle Class")
```

```
plt.ylabel("Average CO₂ Emissions")
plt.xticks(rotation=60)
plt.show()

vehicle_emissions.head(10)
```



Out[16]:
```
Vehicle Class
VAN - PASSENGER             397.212121
VAN - CARGO                 361.500000
SUV - STANDARD              304.836735
PICKUP TRUCK - STANDARD     301.513011
PICKUP TRUCK - SMALL        278.968553
TWO-SEATER                  277.454348
FULL-SIZE                   263.316119
MINIVAN                     262.312500
SUBCOMPACT                  246.448845
STATION WAGON - MID-SIZE    238.698113
Name: CO2 Emissions(g/km), dtype: float64
```

7. Fuel Type Comparison

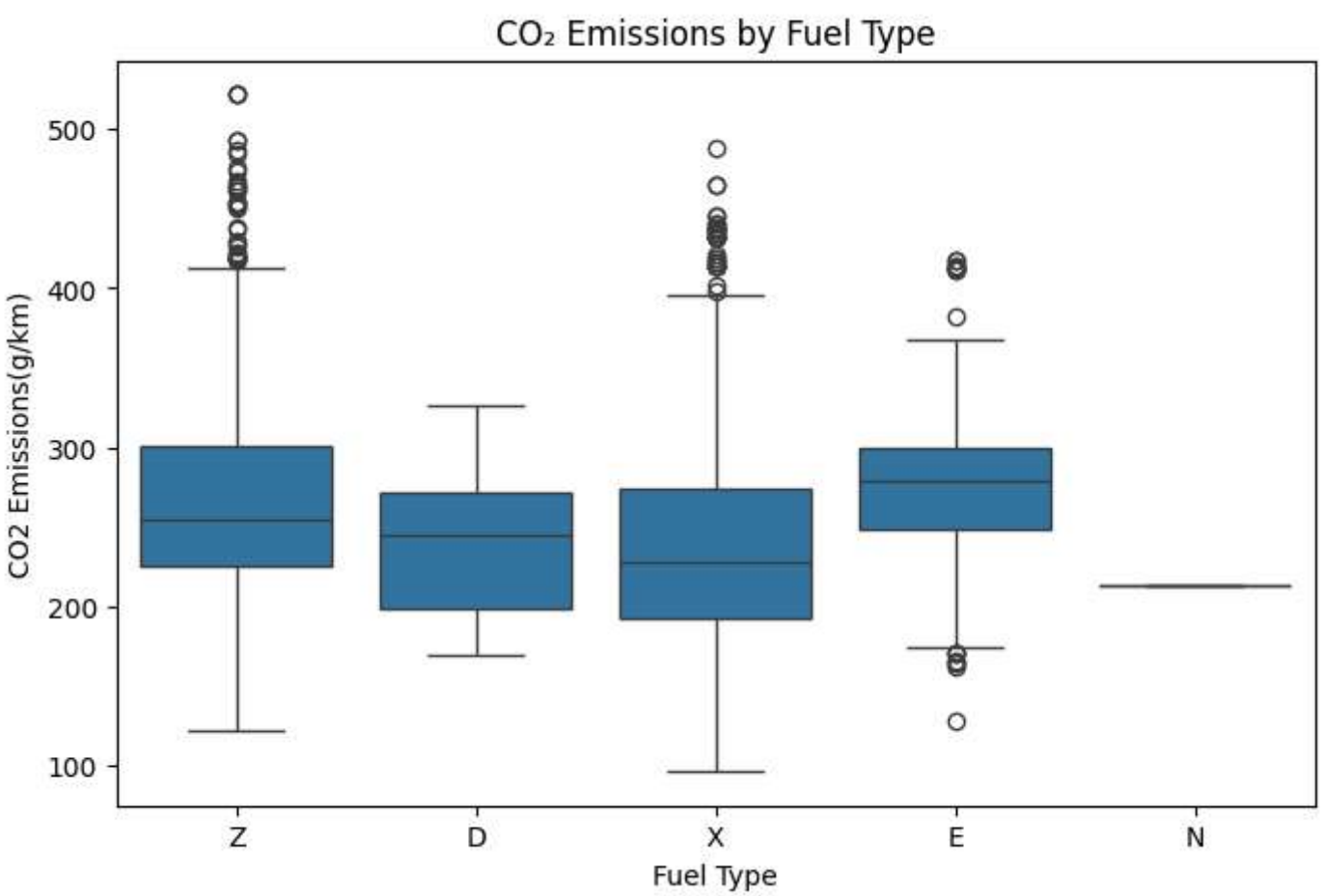
Different fuel types may produce different levels of CO_2 emissions.

This analysis investigates how emissions vary across fuel categories.

In [17]:
```
plt.figure(figsize=(8,5))

sns.boxplot(
    data=df,
    x="Fuel Type",
    y="CO2 Emissions(g/km)"
)

plt.title("CO₂ Emissions by Fuel Type")
plt.show()
```



In [18]:
```
df.groupby("Fuel Type")["CO2 Emissions(g/km)"].mean().sort_values(ascending=False)
```

Out[18]:
```
Fuel Type
E    275.091892
Z    266.043410
D    237.548571
X    235.119329
N    213.000000
Name: CO2 Emissions(g/km), dtype: float64
```

8. Outlier Investigation

Outliers represent vehicles with unusually high or low emissions.

Such observations may correspond to high-performance vehicles, large SUVs, or rare fuel configurations.

In [19]:
```
Q1 = df["CO2 Emissions(g/km)"].quantile(0.25)
Q3 = df["CO2 Emissions(g/km)"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[
    (df["CO2 Emissions(g/km)"] < lower) |
    (df["CO2 Emissions(g/km)"] > upper)
]

print("Number of Outliers :", len(outliers))

outliers.head()
```

```
Number of Outliers : 80
```

Out[19]:

	Make	Model	Vehicle Class	Engine Size(L)	Cylinders	Transmission	Fuel Type	Fuel Consumption City (L/100 km)	Fuel Consumption Hwy (L/100 km)	Fuel Consumption Comb (L/100 km)	Fuel Consumption Comb (mpg)	Er
48	AUDI	R8	TWO-SEATER	5.2	10	M6	Z	21.1	13.8	17.8	16	
52	AUDI	R8 SPYDER	TWO-SEATER	5.2	10	M6	Z	21.1	13.8	17.8	16	
75	BENTLEY	MULSANNE	MID-SIZE	6.8	8	AS8	Z	22.3	14.9	19.0	15	
223	CHEVROLET	EXPRESS 2500 PASSENGER	VAN - PASSENGER	4.8	8	A6	X	21.1	14.2	18.0	16	
224	CHEVROLET	EXPRESS 2500 PASSENGER	VAN - PASSENGER	6.0	8	A6	X	21.9	15.0	18.8	15	

9. Multicollinearity Analysis

Several fuel consumption variables describe nearly identical information.

Variance Inflation Factor (VIF) is used to quantify multicollinearity and identify redundant variables before model development.

In [22]:
```
from statsmodels.stats.outliers_influence import variance_inflation_factor
```

In [23]:
```
vif_features = [
    "Engine Size(L)",
    "Cylinders",
    "Fuel Consumption City (L/100 km)",
    "Fuel Consumption Hwy (L/100 km)",
    "Fuel Consumption Comb (L/100 km)",
    "Fuel Consumption Comb (mpg)"
]

X_vif = df[vif_features]

vif_df = pd.DataFrame()

vif_df["Feature"] = X_vif.columns

vif_df["VIF"] = [
    variance_inflation_factor(X_vif.values, i)
    for i in range(X_vif.shape[1])
]

vif_df.sort_values("VIF", ascending=False)
```

Out[23]:

	Feature	VIF
4	Fuel Consumption Comb (L/100 km)	71236.232743
2	Fuel Consumption City (L/100 km)	28573.049517
3	Fuel Consumption Hwy (L/100 km)	9926.943589
1	Cylinders	79.205365
0	Engine Size(L)	54.498729
5	Fuel Consumption Comb (mpg)	7.025695

10. Feature Selection

The VIF analysis revealed severe multicollinearity among fuel consumption variables.

To avoid unstable regression coefficients and improve interpretability, redundant fuel consumption features were removed.

Retained numerical variables:

- Engine Size(L)
- Cylinders
- Fuel Consumption Comb (L/100 km)

Categorical variables:

- Make
- Vehicle Class
- Transmission
- Fuel Type

In [24]:
```
model_df = df.copy()

drop_cols = [
    "Fuel Consumption City (L/100 km)",
    "Fuel Consumption Hwy (L/100 km)",
    "Fuel Consumption Comb (mpg)"
]

model_df = model_df.drop(columns=drop_cols)

model_df.head()
```

Out[24]:

	Make	Model	Vehicle Class	Engine Size(L)	Cylinders	Transmission	Fuel Type	Fuel Consumption Comb (L/100 km)	CO2 Emissions(g/km)
0	ACURA	ILX	COMPACT	2.0	4	AS5	Z	8.5	196
1	ACURA	ILX	COMPACT	2.4	4	M6	Z	9.6	221
2	ACURA	ILX HYBRID	COMPACT	1.5	4	AV7	Z	5.9	136
3	ACURA	MDX 4WD	SUV - SMALL	3.5	6	AS6	Z	11.1	255
4	ACURA	RDX AWD	SUV - SMALL	3.5	6	AS6	Z	10.6	244

11. Data Preparation

Categorical variables are transformed using One-Hot Encoding.

The dataset is then split into training and testing subsets to evaluate model performance on unseen data.

In [25]:
```
from sklearn.model_selection import train_test_split

X = model_df.drop("CO2 Emissions(g/km)", axis=1)

y = model_df["CO2 Emissions(g/km)"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

print(X_train.shape)
print(X_test.shape)
```

```
(5908, 8)
(1477, 8)
```

In [26]:
```
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder

categorical_cols = [
    "Make",
    "Model",
    "Vehicle Class",
    "Transmission",
    "Fuel Type"
]

numerical_cols = [
    "Engine Size(L)",
    "Cylinders",
    "Fuel Consumption Comb (L/100 km)"
]

preprocessor = ColumnTransformer(
    transformers=[
        (
            "cat",
```

```
            OneHotEncoder(handle_unknown="ignore"),
            categorical_cols
        ),
        (
            "num",
            "passthrough",
            numerical_cols
        )
    ]
)
```

12. Linear Regression Model

A linear regression model is used because:

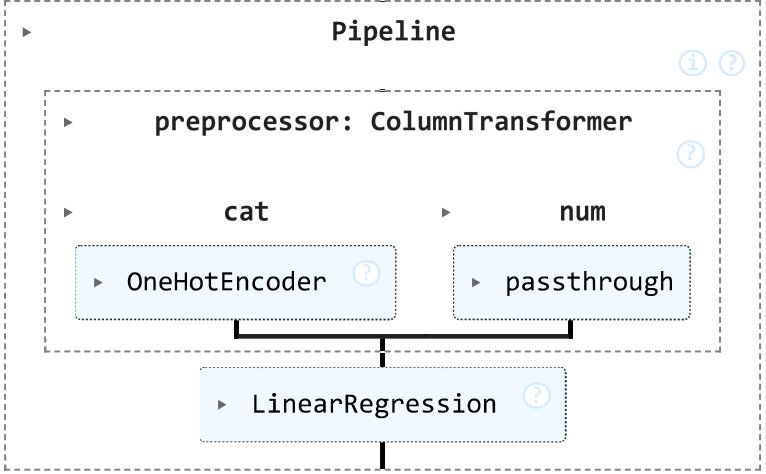
- It is interpretable.
- The target variable is continuous.
- Relationships observed during EDA are largely linear.
- Coefficients can provide insights into emission drivers.

In [27]:
```
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LinearRegression

model = Pipeline(
    steps=[
        ("preprocessor", preprocessor),
        ("regressor", LinearRegression())
    ]
)

model.fit(X_train, y_train)
```

Out[27]:



In [28]:
```
y_pred = model.predict(X_test)
```

13. Model Evaluation

Model performance is evaluated using:

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- R^2 Score

Together these metrics provide a balanced assessment of prediction quality.

In [29]:
```
from sklearn.metrics import (
    mean_absolute_error,
    mean_squared_error,
    r2_score
)

import numpy as np

mae = mean_absolute_error(y_test, y_pred)

rmse = np.sqrt(
    mean_squared_error(y_test, y_pred)
)

r2 = r2_score(y_test, y_pred)

print("MAE :", round(mae,2))
```

```
print("RMSE:", round(rmse,2))
print("R²  :", round(r2,4))
```

```
MAE : 3.36
RMSE: 6.15
R²  : 0.989
```

In [30]:
```
from sklearn.model_selection import cross_val_score

cv_r2 = cross_val_score(
    model,
    X,
    y,
    cv=5,
    scoring="r2"
)

print("Cross Validation R² Scores")
print(cv_r2)

print("Average CV R²:", cv_r2.mean())
```

```
Cross Validation R² Scores
[0.98253395 0.99160338 0.99348842 0.99413015 0.99279401]
Average CV R²: 0.99090998162867
```

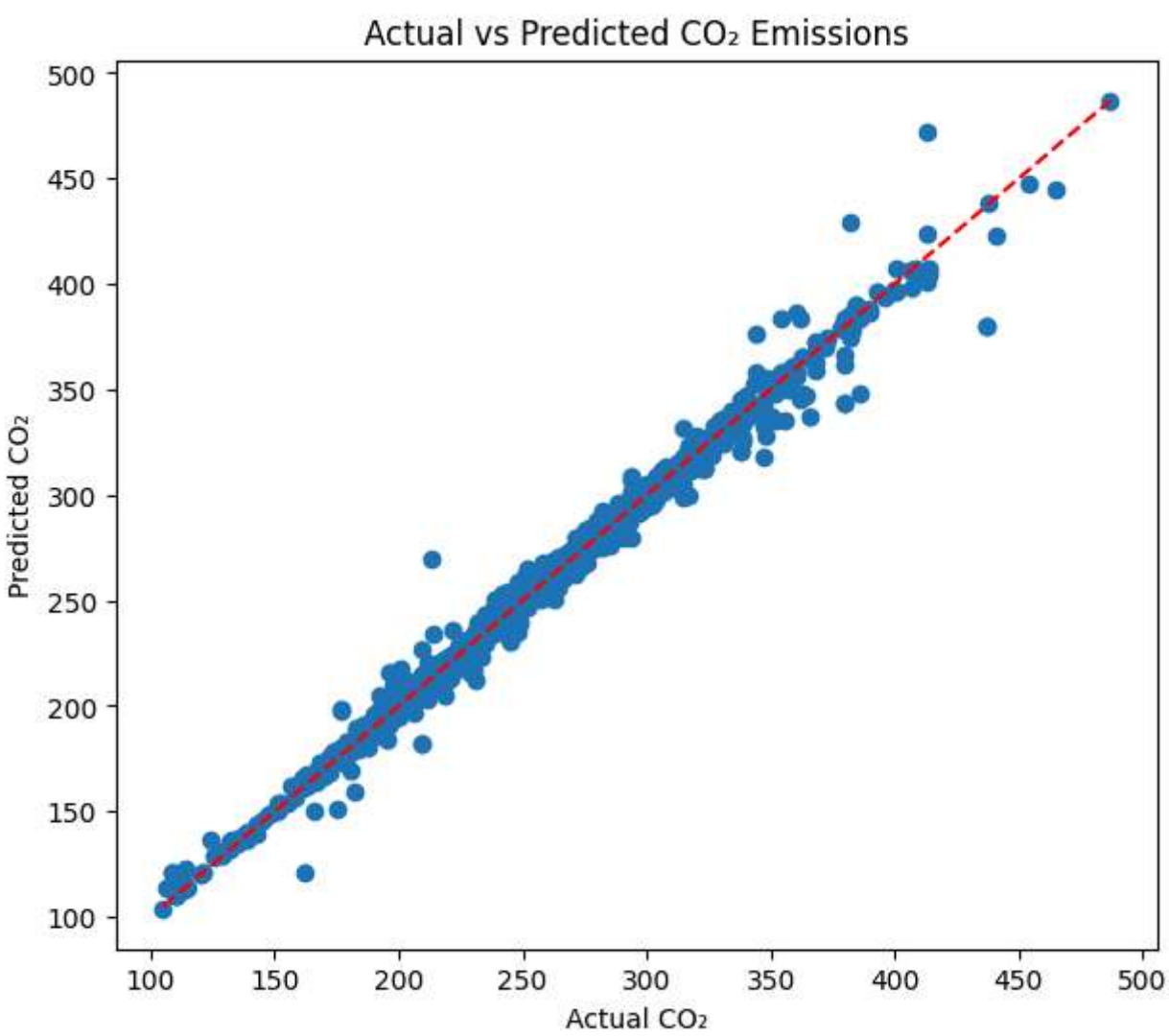
In [31]:
```
plt.figure(figsize=(7,6))

plt.scatter(y_test, y_pred)

plt.plot(
    [y_test.min(), y_test.max()],
    [y_test.min(), y_test.max()],
    'r--'
)

plt.xlabel("Actual $CO_2$")
plt.ylabel("Predicted $CO_2$")
plt.title("Actual vs Predicted $CO_2$ Emissions")

plt.show()
```



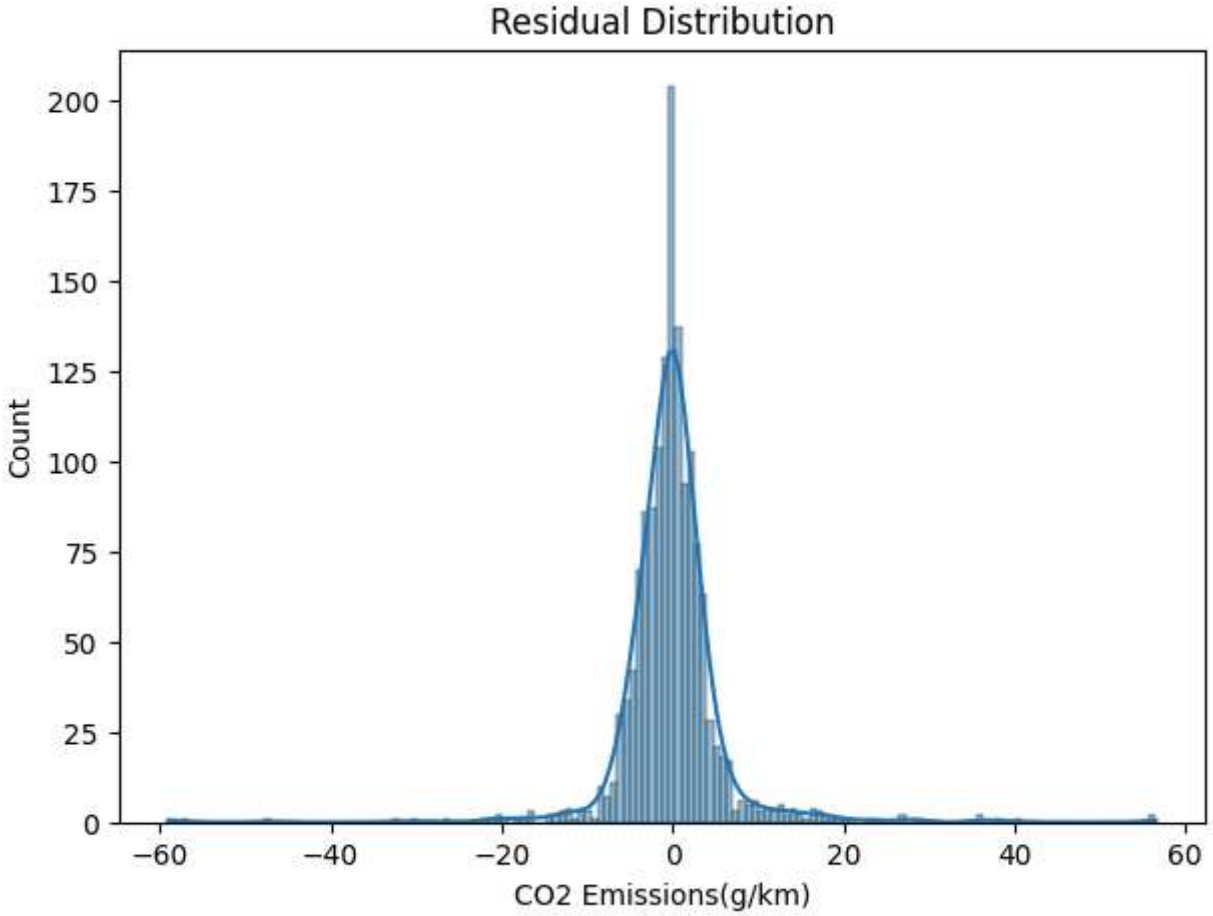
In [32]:
```
residuals = y_test - y_pred

plt.figure(figsize=(7,5))

sns.histplot(residuals, kde=True)

plt.title("Residual Distribution")

plt.show()
```



14. Key Findings

Major Drivers of CO_2 Emissions

The analysis revealed that fuel consumption is the strongest predictor of vehicle CO_2 emissions.

Important influencing factors include:

1. Fuel Consumption Comb (L/100 km)
2. Fuel Consumption City (L/100 km)
3. Engine Size (L)
4. Number of Cylinders
5. Vehicle Class

Vehicles with larger engines and higher fuel consumption generally produce substantially higher emissions.

Vehicle Category Insights

Vehicle classes such as Passenger Vans, Cargo Vans, and Standard SUVs exhibit the highest average emissions.

Smaller vehicle categories such as Compact Cars and Small Station Wagons tend to produce significantly lower emissions.

Fuel Type Insights

Natural Gas vehicles recorded the lowest average emissions in the dataset.

Ethanol and Premium Gasoline vehicles displayed the highest average emissions.

Multicollinearity Findings

Fuel consumption variables were found to be highly correlated with one another.

Variance Inflation Factor (VIF) analysis confirmed severe multicollinearity, requiring removal of redundant variables before model development.

Retaining only the combined fuel consumption metric improved model interpretability and stability.

Outlier Analysis

A small number of high-emission vehicles were identified.

Inspection suggested that these observations correspond to legitimate high-performance vehicles, large vans, and luxury automobiles rather than data errors.

Therefore, outliers were retained during modeling.

15. Model Evaluation Summary

A Linear Regression model was developed using vehicle specifications, fuel characteristics, and categorical vehicle information.

Performance Results

Metric	Value
MAE	3.36
RMSE	6.15

Metric	Value
R^2	0.989
Average CV R^2	0.991

Interpretation

The model achieved extremely strong predictive performance.

The high R^2 value indicates that the selected vehicle characteristics explain nearly all variation in CO_2 emissions.

The low MAE and RMSE values suggest that prediction errors are small in practical terms.

Cross-validation scores were consistent with test performance, demonstrating good generalization and minimal overfitting.

16. Business Recommendations

Based on the analysis, the following recommendations can support emission reduction efforts:

Improve Fuel Efficiency

Fuel consumption is the strongest driver of emissions. Policies encouraging improved fuel efficiency can have the greatest impact on reducing CO_2 output.

Promote Smaller Engine Configurations

Vehicles with larger engines and more cylinders consistently produce higher emissions. Encouraging downsized engines can reduce environmental impact.

Encourage Low-Emission Vehicle Classes

Compact vehicles and smaller station wagons exhibit substantially lower emissions compared to large vans and SUVs.

Support Alternative Fuel Technologies

Natural Gas vehicles demonstrated lower average emissions. Further investment in cleaner fuel technologies may contribute to long-term emission reduction goals.

Regulatory Monitoring

Manufacturers producing high-emission vehicle categories should be encouraged to adopt efficiency improvements and cleaner technologies.

Case Study Question Answers

Q1. Dataset Familiarization

The dataset contains vehicle specifications, engine characteristics, transmission details, fuel types, fuel consumption metrics, and CO_2 emission values. These variables collectively influence vehicle efficiency and environmental impact.

Q2. Data Quality Assessment

The dataset was examined for missing values, duplicate records, and data type inconsistencies. No significant data quality issues were observed. Outliers were investigated and retained because they represented legitimate vehicle categories.

Q3. Feature Relationships

Correlation analysis revealed that fuel consumption metrics, engine size, and cylinder count exhibit the strongest relationships with CO_2 emissions.

Q4. Numerical Variable Trends

Scatterplot analysis showed strong positive relationships between fuel consumption, engine size, cylinders, and CO_2 emissions. Higher values generally correspond to increased emissions.

Q5. Vehicle and Fuel Category Comparison

Passenger Vans, Cargo Vans, and Standard SUVs recorded the highest average emissions. Natural Gas vehicles showed the lowest average emissions, while Ethanol and Premium Gasoline vehicles exhibited higher emission levels.

Q6. Outlier Investigation

Approximately 80 emission outliers were identified. Most corresponded to high-performance sports cars, luxury vehicles, and large commercial vans. These observations were retained because they represent valid real-world vehicles.

Q7. Data Preparation

Categorical variables were encoded using One-Hot Encoding. Multicollinearity among fuel consumption variables was addressed through VIF analysis and redundant variables were removed.

Q8. Predictive Modeling

A Linear Regression model was developed to estimate CO_2 emissions using vehicle specifications, fuel information, and consumption metrics.

Q9. Model Performance

The model achieved excellent predictive performance with:

- MAE = 3.36
- RMSE = 6.15
- R^2 = 0.989
- Average Cross-Validation R^2 = 0.991

These results indicate highly reliable emission predictions.

Q10. Final Conclusions

Fuel consumption is the strongest determinant of CO_2 emissions. Engine size, cylinder count, fuel type, and vehicle class also contribute significantly. Improving vehicle efficiency, promoting smaller engine designs, and supporting cleaner fuel technologies can substantially reduce automotive emissions.

In [ ]: